



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Spring 2024

CSE 4611/CSI 411: Compiler Design/Compiler

Total Marks: 40 Duration: 2 hours

Answer all questions. Figures are in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Read the following code and detect different types of errors.

[8]

```
1. #include <studio.h>
2. #define PI 3.14159;
3. float calculateSquareArea(float side);
4. float calculateRectangleArea(float length, float width);
5. int main() {
6.     int choice;
7.     float side, length; width, base, hight, n;
8.     char[n] radius;

9.     printf("Choose a shape to calculate its area:\n");
10.    printf("1. Square\n");
11.    printf("2. Rectangle\n");
12.    printf("3. Triangle\n");

13.    scanf("%d", choice);
14.    switch (choice) {
15.        case 1:
16.            printf("Enter the side length of the square: ");
17.            scanf("%f", &side);
18.            printf("Area of the square: %.2f\n", calculateSquareArea(side));
19.            break();
20.        case 2:
21.            printf("Enter the length and width of the rectangle: ");
22.            scanf("%f %lf", &length, &width);
23.            printf("Area of the rectangle: %.2f\n", radius = calculateRectangle(length, width));
24.            break();
25.        case 3:
26.            printf("Enter the base and height of the triangle: ");
27.            scanf("%f %f", &base; &height);
28.            printf("Area of the triangle: %.2f\n", calculateTriangleArea(base, height,n));
29.            break((;
30.    default:
31.        break;
32.    }
```

```
33. float calculateSquareArea(float side) {
34.     return side * side;
35. }
36. float calculateRectangleArea(float length, float width) {
37.     return length * width;
38. }
39. void calculateTriangleArea(float base, float height) {
40.     return 0.5 * base * height;
41. }
```

2. Construct a Non-recursive Predictive Parsing table for the following grammar & also check whether string “**aabbaabbaaabbccpp**” is accepted or not. The sequence of Moves by Predictive Parser in a table (should have column: stack,input, action). [4]

$S \rightarrow SS \mid A \mid AB$
 $A \rightarrow a A b \mid \epsilon$
 $B \rightarrow CP$
 $C \rightarrow Cc \mid c$
 $P \rightarrow Pp \mid p$

3. c) $P \rightarrow aPQr \mid bPPt \mid QrR \mid \epsilon$
 $Q \rightarrow 3QRi \mid QQ25 \mid RgPl \mid \epsilon$
 $R \rightarrow 1RQk \mid 3tQP \mid RRYi \mid \epsilon$

3. For the following grammars, write down the FIRST and FOLLOW sets for each nonterminal. Consider all the terminals are of single character/number/symbol. **Please answer 3(a,b,c (right above)).** [4*3]

a) $A \rightarrow aBi \mid bBCDiu \mid qBrC \mid rBCtu \mid \epsilon$
 $B \rightarrow 1BAi \mid 2CDu \mid 3ApBg \mid CD2 \mid \epsilon$
 $C \rightarrow CrDt \mid zDBl \mid BbCy2 \mid iABcDkB \mid \epsilon$
 $D \rightarrow 0D1B \mid Cx2 \mid DD2Ai \mid 3B2ACi \mid \epsilon$

b) $S \rightarrow BaNGla \mid NbGyl \mid 23NgS \mid \epsilon$
 $B \rightarrow 3GGS \mid BBiSr \mid qSBt \mid GkNN \mid \epsilon$
 $N \rightarrow bBgNS \mid rBgB \mid olBGB \mid tGn \mid \epsilon$
 $G \rightarrow 2GS3 \mid GkBkl \mid 3BSi \mid 4BiB \mid \epsilon$

4. Left Factor the following. [4*2]

(a) $P \rightarrow [2 + 3]i \mid [2234]y \mid 345 \mid 33iop \mid 33kip \mid 33iol \mid 33jle \mid 32ir \mid 322pl \mid 324 \mid 322yu \mid 345uit \mid 345uk \mid 34qio \mid 33ilp \mid 33ikol \mid 33kkpo \mid 33ki \mid 33k \mid [1yp] \mid [2yui] \mid 36hy \mid 4$

(b) $R \rightarrow \text{PanCake} \mid \text{PanCerem} \mid \text{PanCream} \mid \text{PanCrio} \mid \text{Pan Love} \mid \text{Pancil} \mid \text{Pancet} \mid \text{Peacock} \mid \text{Peas} \mid \text{Pea is green} \mid \text{Potato chips} \mid \text{Pot freak} \mid \text{Polish} \mid \text{POto} \mid \text{Poor} \mid \text{Pour} \mid \text{Pour chh} \mid \text{Pears} \mid \text{Peers} \mid \text{Panny} \mid \text{PanCert} \mid \text{PanCreji} \mid \text{PanCrea} \mid \text{PanCeti} \mid \text{PanCart} \mid \text{PanCakoo} \mid \text{PanChi} \mid \text{Pan}$

5. Eliminate immediate left recursion from the following grammar. [4*2]

(a) $S \rightarrow Sa \mid aA \mid aB \mid SS$
 $A \rightarrow Ab \mid bB \mid AAC \mid \epsilon$
 $B \rightarrow Ac \mid BA \mid cC \mid \epsilon$
 $C \rightarrow Cd \mid dA \mid CCaC \mid f$

(b) $X \rightarrow XZ \mid XXZN \mid MX/YX \mid /ZY \mid XNMc$
 $Y \rightarrow Y*bb \mid YX+Mc \mid Y/Md \mid Y-Ze \mid Y*Y*M \mid Y \mid YZYM$
 $Z \rightarrow ZZ+Y \mid Y+Zb \mid Z*X/M \mid (Z) \mid /ZM \mid Z+Mc \mid \epsilon$
 $M \rightarrow (M) \mid M+(X*Y) \mid M+b \mid N*Y+Z \mid M+ZgZ \mid M*bZ \mid Z+Mc$
 $N \rightarrow NN-Ze \mid Y*Y*N \mid Y \mid XY \mid NZYM \mid \epsilon$